work groups, in contrast to other kinds, those who worked in close association with one another had different skills and consequently different economic (union) interests. In addition, these workers were more closely tied to their work, because of its interdependent nature, and could not move about in the shop to associate with workers of like interests as could, for example, maintenance workers. But union solidarity requires that workers with the same economic interests have opportunities for freely associating with one another.

The complex and often contradictory findings with regard to the conditions that produce group cohesiveness and social solidarity suggest that much research remains to be done in this area. The results will probably show that the conditions of group solidarity are much more complex than is implied in the simple hypothesis that solidarity depends directly on membership homogeneity, or on prestigeful characteristics, or even on the importance of the group's function for the larger organization.

Comparison of Two Formal Organizations. It has long been asserted that work groups should not be studied in isolation but in the context of the larger organization of which they are a part.[59] However, few studies have used a comparative approach to show how specific differences between formal organizations influence work group structure. The data provided by our studies of the two welfare agencies make such a comparison possible. A discussion of the main characteristics of these agencies appears in the Appendix. Here we shall only summarize some of the main differences between the two formal structures. City Agency was considerably larger than County Agency, the former employing over 1000 persons, the latter, less than 200. The management of County Agency tended to place more emphasis on social-work training than did that of City Agency. There was a manual of operating rules for each worker in County Agency but not in City Agency. City Agency had considerably more turnover than County, and thus County employees enjoyed higher average seniority. Finally, the office ecology was not the same. Although in both organizations members of several units occupied the same work room, members of the same unit had adjoining desks in City Agency but not in County, where members of all units were scattered throughout the room. We shall explore some of the implications of these differences for social relations within the two agencies.

First, there was a difference in orientation to clients. Although the modal orientations of the caseworkers in the two agencies were quite similar, the extremes were different. For example, the proportion who

[59] For one statement calling for such an approach, see William F. Whyte, "Small Groups and Large Organizations," John H. Rohrer and Muzafer Sherif (eds.), *Social Psychology at the Crossroads,* New York: Harper, 1951, pp. 299–312.

firmly favored raising the assistance allowance was about the same in City Agency (48 per cent) as in County Agency (56 per cent), but while fully 18 per cent of the City workers expressed the opposite opinion—that grants were sufficient and should not be raised—only 1 per cent of the County workers did. Similarly, whereas most employees in both agencies felt that clients were basically honest, 12 per cent of the City workers held that many clients were dishonest while only 1 per cent of those at County did. These findings suggest that one function of a professionally oriented management, such as that in County Agency, is to screen and eliminate those workers with the most negative attitudes toward clients. Another factor that may well have contributed to these differences, however, was the lower turnover rate at County Agency, which enabled County management and supervisors to know and train their workers better than was possible in City Agency. County management could also better afford to weed out poorly qualified workers (such as those hostile toward clients) than City management with its perennial staff shortage.

We expected that the fact that members of a unit sat together at adjacent desks in City Agency but not in County Agency would provide more opportunity for intra-unit social interaction and result in more cohesive units in the former agency. This prediction was not borne out. When cohesion was measured by the proportion of ingroup friendship choices in a unit, the medians for the two agencies were similar: .19 for the twelve groups studied in City Agency and .22 for the eleven in County Agency. Moreover, the mean proportion of ingroup choices was higher in County Agency (.29) than in City (.18), as was the standard deviation (.24 and .09 respectively). Inspection of the unit scores revealed that only one unit in City Agency had a score above .30 but four units in County Agency were well above this figure. Thus, while both agencies had several units with relatively low cohesion, County Agency but not City Agency had several units with very high cohesion. It appears that ecological factors exerted little influence on ingroup friendship choices; instead, the relatively low turnover rate of workers in County Agency may well have been the decisive factor that explains the differences between the agencies. High turnover in City Agency prevented any work group from developing really high cohesion whereas low turnover in County Agency allowed some work groups to do so despite the scattered seating arrangement, although most groups did not, perhaps in part because of the office ecology.

Another difference observed between the two agencies was in the patterns of consultation among workers. In County Agency, workers who said they worried about their cases were more involved in the consultation network. They were more often consulted and, particularly,

they were more apt to consult others. Thus, 39 per cent of the 18 high worriers, 15 per cent of the 48 medium worriers, and none of the 20 nonworriers reported that they consulted colleagues often.[60] In City Agency, on the other hand, worriers did not participate more in consultation than others. This difference may well have been due to the existence of a procedure manual in County Agency and its absence in City Agency. Consultation among colleagues is of great importance not only for obtaining information but also for social support that facilitates decision-making, as we shall point out in Chapter V. Most caseworkers in both agencies regularly discussed problems with colleagues. The absence of a procedure manual accentuates the need for informal consultation, especially for inexperienced workers, who must often ask colleagues or the supervisor what the official procedures are, whereas inexperienced workers can look up this type of information themselves if a manual is available. In City Agency, therefore, some workers had good reason to ask disproportionately many questions, whether they were particularly anxious about their work or not. But since the procedure manual in County Agency obviated this objective need for advice of some workers, the subjective need for advice experienced by workers primarily governed their participation in consultation, and this subjective need was greatest for anxious worriers who wanted to make doubly sure of their decisions before taking action.

This interpretation implies that inexperienced workers consult more and experienced ones are more often consulted in the agency without a manual, but that there are no such differences in the agency with a manual. The findings confirm this inference. In City Agency 78 per cent of the 18 oldtimers but only 50 per cent of the 42 less experienced workers were regularly consulted by colleagues, whereas in County Agency experience did not affect being consulted: 44 per cent of the 48 oldtimers and 43 per cent of the 37 others were regularly consulted. Frequency of consulting reveals complementary patterns. The absence of a rule book apparently puts a premium on experience. It seems that written procedures can become a functional substitute for personal experience, and that this aspect of bureaucratization affects informal relations in a formal organization in subtle ways by altering the characteristics of individuals that govern their informal status in the peer group.

Whereas seniority affected consultation in City but not in County Agency, it was associated with popularity in County but not in City Agency. In County Agency, 62 per cent of the oldtimers and 38 per cent of the workers with less than three years' service were chosen as

[60] Formal observation records also substantiate this pattern: 55 per cent of 11 high worriers, 31 per cent of 29 medium worriers, and 27 per cent of 11 nonworriers were found to be high on initiated contacts per hour to fellow workers.

"friends" by several colleagues. In City Agency, exactly 50 per cent of each group were popular by this criterion. These differences may be due to differences in outside opportunities for employment and in relative turnover, and thus in the process of selection. In the metropolis in which City Agency was located there were many private social-work agencies with superior standards of professional casework. This condition did not hold for the community in which County Agency was located: there were few private agencies, and none whose professional standards were superior to those at County Agency. Besides, other employment opportunities appropriate for college graduates were probably greater in the larger city; for example, there were undoubtedly more opportunities for employment in a variety of white-collar jobs in the metropolis than in the smaller community in which County Agency was located. As a result, some of the best workers in City Agency left for other jobs. Workers who were ambitious, versatile, and oriented toward professional social work often sought other employment offering better pay and more professional opportunities. However, the best workers in County Agency tended to remain, since there were no important competitors for the abilities that they had to offer. The process of negative selection that occurred in City Agency probably made old-timers there somewhat less desirable companions, while in County Agency the better workers and those most committed to social work stayed on and enjoyed high popularity.[61]

In this section, we have focused on the *differences* in the work-group structures of the two agencies. It should be noted, however, that many patterns were the same in both agencies. For example, seniority was directly associated with being respected by colleagues and with being judged competent by the supervisor in both agencies; experience promotes proficiency whether there is a procedure manual or not. But when there is no manual, experience enhances a person's chances of being consulted quite independently of his proficiency. If respect is held constant, seniority and being often consulted were still directly related in City Agency (but not in County Agency). In both agencies, consultants were more popular than workers rarely consulted. And popular workers in both agencies were more self-confident than others (stating that they could do their work without aid from the supervisor).

The foregoing comparison of two formal organizations has provided insights that could not have been derived from the study of a single case. For instance, our conclusion that a written procedure manual influences social relations in certain ways depends on data from two contrasting organizations. Since these interpretations are based on only

[61] This interpretation assumes that the best caseworkers are typically most popular among colleagues. The data confirm this assumption: there was a direct relationship between being considered one of the best caseworkers and popularity in both agencies.

two cases, however, we cannot even tentatively test our explanatory hypotheses about the influences of given aspects of the formal organization. Differences other than those noted may have been responsible for the observed differences in social relations and group structures. A comparison of many formal organizations, or at least segments of organizations, is required to determine which one of many possible institutional characteristics actually has brought about an observed result.

Concluding Remarks

This chapter has been devoted to an examination of the social structure of work groups. We have discussed various aspects of informal organization, including norms, cohesion, and informal status, and have analyzed some of the effects exerted by group structure. In the final part dealing with some of the determinants of group structure, we explored first the influence of background characteristics of group members and then those of the larger formal organization. Another major factor influencing the structure of work groups is supervision. It may be thought of as the connecting link between the formal organization and the work group. Supervision will be analyzed in Chapter VI, after we have discussed processes of communication in organizations.

CHAPTER FIVE

Processes of Communication

In preceding chapters we have spoken much of social interaction but little of communication, although when people interact they communicate meaningful information. The two terms refer to the same social processes but to different aspects of them. The concept of social interaction focuses principally upon the formal characteristics of social reciprocity indicate its dimensions. The concept of communication, on the other hand, directs attention to the meaningful content conveyed in the encounter, and its characteristics are described by such terms as flow of messages, obstacles, positive and negative reactions, and exchange.

In this chapter we shall draw on both experimental and field studies to analyze the significance of communication processes for group life and particularly for the performance of tasks. Interaction as well as communication aspects of these processes will inevitably be involved in the analysis. Our procedure will be to attempt to interpret contrasting findings; we shall examine some cases and formulate tentative generalizations, then search for negative cases and refine our interpretation accordingly. The major focus will be upon communication among colleagues. Starting with the analysis of communication processes among peers, we shall turn to the significance of status differences for these processes. In a sense, this chapter provides a transition between the discussion of peer relations in Chapter IV and the analysis of hierarchical relations in Chapter VI.

Experiments on Communication and Performance

The Significance of Social Interaction. A number of experiments have compared the task performance of individuals and groups; most have found groups to be superior. An early experiment conducted by Allport on the speed with which subjects were able to produce free associations from a word stimulus contrasted subjects who worked alone with others who worked in a room where several were engaged in the

same task.[1] Subjects were instructed not to compete. The performance in the social setting was found to be superior to that in isolation. Allport's analysis suggests that, instructions notwithstanding, the social situation led to rivalry. The competitive element introduced by seeing how fast others worked was the factor responsible for the superior performance of individuals in the social setting. Another experiment, reported by Taylor and Faust, compared the performance of single individuals, of pairs working together, and of groups of four working together on finding correct solutions to problems which consisted of a standardized version of the parlor game "twenty questions."[2] Pairs and groups of four were found to be superior to individuals working alone, whether number of questions used, amount of time needed for reaching a correct solution, or number of failures was the criterion of performance. No significant differences in performance were found between the two-man and the four-man groups. Although both experiments were concerned with the effect of social factors on performance, they differed in the operational definition of "social": Allport examined the performance of individuals *working separately in the presence of others,* while Taylor and Faust studied the performance of several persons *working together.*

A third experiment allows us to isolate the effects of working together from the influence of simply being in a social situation. Perlmutter and Montmollin conducted an experiment in which 20 groups of three individuals each were asked to recall correctly a list of nonsense syllables previously presented.[3] In the first part of the experiment, half the subjects worked individually, the answers of each being scored separately, while the other half collaborated on a composite group score; in the second part the situation was reversed so that those who had previously worked as individuals collaborated and those who had previously collaborated in groups worked individually. Even when subjects worked as individuals, however, they were in the presence of others engaged in the same task. Hence, this design eliminates the effect of the social situation, the focus of Allport's study, and isolates that of collaboration. The results show that the performance of groups was, on the average, superior to that of individuals. Besides, the individuals who began by working in a group performed better later, when working on their own, than did those who did not have previous group experience. It appears that collaboration with others

1 Floyd H. Allport, "The Influence of the Group upon Association and Thought," *Journal of Experimental Psychology,* 3 (1920), pp. 159–182.

2 Donald W. Taylor and William L. Faust, "Twenty Questions," *Journal of Experimental Psychology,* 44 (1952), pp. 360–368.

3 Howard V. Perlmutter and Germaine de Montmollin, "Group Learning of Nonsense Syllables," *Journal of Abnormal and Social Psychology,* 42 (1952), pp. 762–769.

in problem-solving improves performance quite independently of any improvements resulting from the mere presence of others, and that the skills acquired in group learning even contribute to subsequent individual work.

What accounts for the superiority of group over individual performance? Is it mere chance? Chance, in the sense of probability, does exert some effect. In the experiment just reported, the chance that a given syllable will be recalled by one of the several persons in a group is greater than the probability that it will be recalled by any single one of them. Studies have found that statistical probability accounts for some of the superiority of groups, but not for all of it.[4] What, then, are the other factors that account for the group's superior performance? An early experiment by Shaw provides one important clue to the answer.[5]

Shaw presented complex puzzles (for example, transporting missionaries and cannibals across a river under specified conditions, or arranging words to complete a poem) to individuals and to groups of four working together. Groups were found to be superior to individuals in solving the puzzles. Shaw also placed an observer in the room where the groups were working and obtained systematic data on the interaction that occurred in the course of trying to solve the problems. Analysis of the interaction data showed that incorrect suggestions were often made and then rejected; occasionally the person who made the suggestion later saw his own error, but more often (in 160 of 217 cases) another person corrected him. This experience seems to indicate that social interaction provides an error-correcting mechanism. It is not easy for an individual to see his own mistakes, for he approaches a problem within a particular framework of perceptions and assumptions. The fact that various individuals bring different frameworks as well as different experiences and knowledge to the group multiplies the chances that an error made will be detected. The error-correcting property of social interaction is one reason why groups are superior to individuals in solving problems.

Not all studies, however, have found group performance to be superior to that of individuals. An examination of those tasks that groups do not perform as well as individuals will allow us to refine our analysis by specifying the conditions under which groups are generally superior. An experiment by Thorndike reports that groups were superior to individuals in *solving* crossword puzzles but that individ-

[4] See, for example, Irving Lorge and Herbert Solomon, "Two Models of Group Behavior in the Solution of Eureka-type Problems," *Psychometrika,* 20 (1955), pp. 139–148.

[5] Marjorie E. Shaw, "A Comparison of Individuals and Small Groups in the Rational Solution of Complex Problems," *American Journal of Psychology,* 44 (1932), pp. 491–504.

uals were superior to groups in *constructing* them.[6] Both these tasks are quite complex. What is the crucial difference between them? In solving puzzles, there is one correct answer and each step taken toward the solution is either correct or incorrect. This description applies not only to solving crossword puzzles but also to most of the other problems utilized in experiments on task performance. On the other hand, in constructing crossword puzzles there is no single correct solution. The problem is not solved by finding the one right word but rather by coordinating early with later steps. Any word will do as long as it fits together with previous selections and does not obstruct later steps. A guiding framework is important for achieving such coordination, and an implicit framework, which is not easily communicated to others, guides individuals in constructing puzzles. The different frames of reference that individuals bring to a group aid in the search for a correct solution among several alternatives but impede the coordination of different ideas into one consistent whole.[7] Thus, we can refine our generalization and derive a principle that can account for apparently contradictory findings: social communication in a group working on a common task provides a battleground of ideas, which stimulates thinking and, particularly, facilitates detecting false leads, but which simultaneously interferes with coordination. Whether the performance of groups is superior or inferior to that of individuals depends, therefore, on whether the essential task requirement is finding the best solution for a problem or achieving effective coordination.

Social interaction not only provides a correcting mechanism for sifting ideas but also contributes to problem-solving in groups by furnishing social support to individual members. Problem-solving engenders anxieties. An individual makes a tentative first step and begins to wonder whether or not it is correct. His uncertainty produces mental blocks which interfere with his further associations and ideas. In a group situation, however, good suggestions are likely to receive the approval of others. Approval may be explicit in the form of verbal comments or may be implicit in the behavior of others as they nod, smile, look expectantly, or just refrain from interrupting. Such social approval mitigates anxieties and frees the individual to continue his train of thought. There seems to be a contradiction between saying that interaction involves error correction and that it involves social support—the rejection and the approval of ideas—but it does both, only for differ-

[6] R. L. Thorndike, "On What Type of Task Will a Group Do Well?" *Journal of Abnormal and Social Psychology,* 33 (1938), pp. 409–413.

[7] See H. G. McCurdy and W. E. Lambert, "The Efficiency of Small Human Groups in the Solution of Problems Requiring Genuine Cooperation," *Journal of Personnel,* 20 (1952), pp. 478–494, for another study where the task is coordination and where individuals were found to be superior to groups.

ent suggestions and for different persons. Since interaction provides a battleground for ideas that serves to sift the suggestions made by the various members of the group, the ideas of some members are rejected, and these members do not experience the support that facilitates their thinking. But the ideas of others, or even ideas of the same members at other times, are approved, and this support helps them to develop their ideas further. Ideally, only wrong suggestions are rejected, and the thinking of those whose early suggestions promise to lead to the correct final solution is facilitated by social approval. But actually, other factors may influence this interaction, causing group members sometimes to reject correct solutions and to accept incorrect ones.

Still another aspect of social interaction that contributes to problem-solving is illustrated by a paradox implicit in the conclusions of Allport and those of the other experimenters discussed. The social situation was usually found to improve performance, but in Allport's experiment this improvement was due to the competition engendered by the presence of others, whereas in most experiments it was the result of cooperation among group members. Competition and cooperation seem to be opposites, but both, nevertheless, apparently improve performance in comparison to what it would be without either kind of social interaction, that is, over the performance of individuals working in isolation. To be sure, we can also ask whether competition or cooperation in a group makes a greater contribution to task performance. Experiments as well as field studies indicate that performance in competitive groups is inferior to that in cooperative groups.[8] In the light of Allport's findings, as well as others, this difference should probably be interpreted to indicate not that competition inherently impedes performance but that some of its by-products do, notably, a lack of social support and an inability to pursue common goals in predominantly competitive groups. (We have seen in Chapter IV how work groups tend to protect themselves against these consequences of competition that threaten their solidarity.) Some forms of competition occur in most groups, if not in all, and contribute to task performance.

In most groups, and particularly in those that appear to be basically cooperative, social interaction contains competitive as well as cooperative elements. Thus, in the experimental groups under consideration, instructions that members were to work together and pursue common objectives gave them an incentive to cooperate and support one another's efforts, since all benefited from the contribution each made to the correct solution of the problem. (In solidary work

[8] For example, see the experiment by Morton Deutsch, "An Experimental Study of the Effects of Cooperation and Competition upon Group Process," *Human Relations,* 2 (1949), pp. 199–231; and the field study by Peter M. Blau, *The Dynamics of Bureaucracy,* Chicago: University of Chicago Press, 1955, pp. 49–67.

groups, too, there is much cooperation in the form of consultation, enforcement of common norms, and the like.) Within this fundamentally cooperative framework, however, interaction will give rise to competitive processes since the members of a group typically seek to gain the respect of one another. Each member wants to be highly thought of by the rest—preferably more highly than others are. Such competition has positive consequences for performance. In groups working together on a task, competition for respect motivates members to make good suggestions and also to prove their superior abilities by criticizing the suggestions of others. This competition is the social mechanism that mobilizes the energies of group members and encourages them to devote effort to finding solutions. It is this social mechanism that induces them to bring their different frameworks to bear on problems, resulting in the correction of errors. Such competition does, of course, interfere with the furnishing of social support to group members. Ideally, the two processes must be kept in balance so that the competition is not allowed to tear the group apart and the supporting processes are not allowed to lull the group into a situation where the members turn from working on task problems to an exclusive concern with one another's feelings.[9]

Dysfunctions of Hierarchical Differentiation. The superiority of groups over individuals in certain kinds of task performance has been attributed primarily to three factors: (1) the sifting of suggestions in social interaction serves as an error-correction mechanism; (2) the social support furnished in interaction facilitates thinking; and (3) the competition among members for respect mobilizes their energies for contributing to the task. If our interpretation is correct, a comparison of groups that differ in these characteristics should reveal appropriate differences in performance. The best test of these explanatory hypotheses would be to introduce experimental differences in each of the three factors while controlling the remaining two and other relevant conditions. In the absence of such experimental evidence, we shall attempt to test our interpretations indirectly by examining social conditions that appear to inhibit these processes and noting the consequences for performance.

[9] Bales has discussed this dilemma posed by, in his terms, "task" and "socio-emotional" problems. He suggests that concentration on the group task gives rise to disturbances in the relations between members and that these socio-emotional problems must be solved if the group is to continue to work successfully on the task. Two mechanisms to cope with this dilemma are differentiation of time (at one period, task problems are worked on, and at another, socio-emotional problems) and differentiation of roles (some members focus on the task and others concentrate on resolving conflicts and tensions). See Robert F. Bales, "The Equilibrium Problem in Small Groups," Talcott Parsons *et al.*, *Working Papers in the Theory of Action,* Glencoe, Ill.: Free Press, 1953, pp. 111–161.

Hierarchical differentiation of status, particularly when formally established, appears to curtail these three group processes. First, explicit status distinctions tend to reduce social interaction and social support. A field study by Wessen of the personnel in a large general hospital found that social interaction typically followed status lines and was inhibited by status boundaries. Most social contacts of doctors were with other doctors, nurses primarily associated with other nurses, and the same was true for lower-status ward personnel.[10] An investigation by Kelley found that in experimentally created hierarchies feelings of friendliness across hierarchical boundaries declined when the upper-status individuals were insecure in their positions and the lower-status individuals had no chance of upward mobility.[11] Several studies report a tendency for lower-status group members to direct their friendship choices disproportionately to upper-status members.[12] Since upper-status members tend not to reciprocate but to direct their choices to others also high in status, lower-status members do not receive their share of the social support that is needed for stimulating thought and making suggestions.

Second, formally instituted status differences tend to undermine the process of competition for respect. In a *peer* group, a member's standing rests primarily on the respect of others, and this fact makes their respect and deference most important. But in the presence of formal distinctions of status, the respect of others is not the primary basis of a person's social standing, and competition for respect loses much, though not all, of its significance. Besides, status distinctions generally destroy the *laissez-faire* character of competition for respect. Members no longer seek to be respected by any and all, inasmuch as being respected by those of high status is of more significance than being respected by others of lower status. A field experiment conducted by Hurwitz and his colleagues provides some support for this interpretation.[13] Participation in discussions at a conference on mental hygiene was found to be affected by the influence and the professional prestige of various members. Records of frequency of communication between conference members reveal that those of high prestige participated most in discussions, and that both

10 Albert F. Wessen, "Hospital Ideology and Communication between Ward Personnel," E. Gartly Jaco (ed.), *Patients, Physicians and Illness,* Glencoe, Ill.: Free Press, 1958, pp. 453–458.

11 Harold H. Kelley, "Communication in Experimentally Created Hierarchies," *Human Relations,* 4 (1951), pp. 39–56.

12 See the studies by Jacob I. Hurwitz *et al.,* "Some Effects of Power on the Relations among Group Members," Dorwin Cartwright and Alvin Zander (eds.), *Group Dynamics,* Evanston, Ill.: Row, Peterson, 1953, pp. 483–492; and Muzafer Sherif, "A Preliminary Experimental Study of Inter-group Relations," John H. Rohrer and Muzafer Sherif (eds.), *Social Psychology at the Crossroads,* New York: Harper, 1951, pp. 406–410.

13 Hurwitz *et al., op. cit.,* pp. 486–489.

low- and high-prestige persons directed most of their remarks to high-prestige persons. Liking choices were disproportionately often directed to high-status individuals, and there seemed to be more concern with receiving liking choices from high- than from low-status members. In short, participants appear to have competed primarily for being regarded and liked by the most prestigeful others. Seeking to win the approval of an elite has undoubtedly implications for conduct quite different from those of striving to earn the respect of most members of a group.

Third, status differences distort the error-correcting function of social interaction. It is not easy to oppose the judgment of a person with superior power or prestige, and most people will think twice before doing so. At the very least, they will want to be quite sure of their criticism before voicing it. In contrast, people do not usually hesitate to criticize the opinion of a person whom they do not respect. Indeed, lack of respect will incline them to find fault with proposals. Hence, there is a reluctance to criticize the suggestions offered by high-status members and a ready inclination to reject the suggestions and criticisms offered by low-status members. This interpretation gains support from an experiment in which Torrance observed the interaction during problem-solving sessions in three-man groups composed of Air Force personnel of different military ranks.[14] He found that the incorrect suggestions of high-status group members (crew captains) were disproportionately often accepted and the correct suggestions of low-status members (sergeant gunners) were disproportionately often rejected. The fact that status influences whose opinion carries most weight interferes with the detection of errors in group problem-solving.

There is, then, some indication that differentiation of hierarchical status in groups attenuates the very characteristics that have been hypothesized to be responsible for the superiority of groups over individuals in problem-solving. The interpretation implies, consequently, that the more pronounced the hierarchical differences in a group, the less effectively it will perform. There is some evidence in support of this inference. For example, in Torrance's experiment, permanent combat crews performed less well in solving problems than did temporary crews composed of similar personnel. Torrance accounts for this finding by explaining: "In the temporary crews, the effects of status differences seems [*sic*] to have been diminished and all members less frequently fail to influence when they have the cor-

[14] E. Paul Torrance, "Some Consequences of Power Differences on Decision Making in Permanent and Temporary Three-Man Groups," A. Paul Hare, *et. al.*, (eds.), *Small Groups,* New York: Knopf, 1955, pp. 482–492.

rect answer."[15] The famous studies on the effects of social climates inspired by Lewin and conducted by White and Lippitt also tend to support our prediction.[16] White and Lippitt conclude that groups of boys under "democratic" leaders performed more "efficiently"—if this term is defined to include member satisfaction as well as achievement of work goals—than those under autocratic leaders. The efficiency was especially evident in the groups' ability to work and cope with problems in the absence of the adult leader. Finally, several studies have reported that workers under authoritarian supervision do not perform as well as those whose supervisors minimize status distinctions by delegating work, encouraging discretion, and similar practices.[17] These studies suggest that, as was expected, a diminution of status differentials improves performance.

Functions of Hierarchical Structure. Research evidence, however, does not entirely support our expectation regarding the adverse effects of hierarchy on performance. Some studies have found groups with a status hierarchy to perform better than those with less differentiation of social status. Let us examine these studies and their implications for our theory. An experiment by Maier and Solem contrasts groups of five or six persons with and without an appointed discussion leader.[18] The task was solving mathematical problems. Groups with a leader proved to be superior in performance to those without one. An analysis of the interaction during the problem-solving sessions provides a clue for understanding this apparently negative case. Majority opinions on how to attack a problem tended to develop in these groups, and the majority often exerted pressures to suppress minority opinions, including correct suggestions. Discussion leaders, who acted as chairmen, saw to it that all members had an opportunity to express their views. The fact that persons in the minority had more opportunity to contribute to the correct solution in groups with a discussion leader than in those without seems to have been responsible for the superior performance of the former.

In this experiment, the social differentiation of a group into a majority and a minority created obstacles to the free flow of commu-

[15] *Ibid.*, p. 485.

[16] Ralph White and Ronald Lippitt, "Leader Behavior and Member Reaction in Three 'Social Climates,'" Cartwright and Zander, *op. cit.*, pp. 585–611. See also the same authors' *Autocracy and Democracy*, New York: Harper, 1960.

[17] See, for example, the study by Robert L. Kahn and Daniel Katz, "Leadership Practices in Relation to Productivity and Morale," Cartwright and Zander, *op. cit.*, pp. 612–628; and Michael Argyle *et al.*, "Supervisory Methods Related to Productivity, Absenteeism, and Labour Turnover," *Human Relations*, 11 (1958), pp. 23–40.

[18] Norman R. F. Maier and Allen R. Solem, "The Contributions of a Discussion Leader to the Quality of Group Thinking," *Human Relations*, 5 (1952), pp. 277–288.

nication, just as hierarchical differentiation did in many other groups, and both forms of differentiation had adverse effects on problem-solving. The presence of a leader who removed obstacles to communication—those created by other aspects of group differentiation—improved the group's problem-solving abilities. This new information allows us to refine our theory by specifying why and under which conditions a status hierarchy interferes with performance. We have seen that the free flow of communication—in the form of criticisms, suggestions, manifestations of respect, and expressions of approval furnishing social support—furthers problem-solving. Hierarchical differentiation of status usually impedes the free flow of communication, and for this specific reason such differentiation tends to be detrimental for task performance. If leadership does not block but frees the flow of communication—as in the case of a discussion leader or procedural chairman in contrast to a dominant power—then leadership will further rather than hinder problem-solving.[19]

Another investigation permits further refinement of our interpretation. Heinicke and Bales studied developmental trends over four sessions of initially leaderless groups engaged in the discussion of a human-relations case.[20] The groups were instructed to arrive at a common solution. Status differences between members (measured by sociometric choices) crystallized in some groups but not in others, and the former were found to be more successful in reaching consensus on case problems. Why did the groups that established a status hierarchy perform the task more successfully? The reason may well be that the task was chiefly one of coordination. The criterion of success was not the quality of the recommendation made but whether the group could agree on a single recommendation. As we noted earlier when comparing the performance of individuals and groups, the free flow of suggestions and criticisms facilitates the sifting of ideas to find the single correct one, but this same process impedes coordination. Hierarchical differentiation of status, by curbing the free flow of ideas, facilitates the coordination of opinions to achieve consensus, just as the complete absence of communication (individual performance) facilitates coordination. In sum, groups are superior to individuals, and groups in which there is a free flow of communication are superior to groups in which differentiation impedes communication, in solving problems which call for a single correct or best answer; but indi-

19 This restatement may well also account for the finding in the White and Lippitt study that groups with "democratic" leaders were more efficient than groups with "*laissez-faire*" leaders. See the descriptions of these leadership styles in White and Lippitt, "Leader Behavior and Member Reaction in Three 'Social Climates,'" *op. cit.*, p. 586.

20 Christoph Heinicke and Robert F. Bales, "Developmental Trends in the Structure of Small Groups," *Sociometry*, 16 (1953), pp. 7–38.

viduals are superior to groups, and hierarchically differentiated groups are superior to undifferentiated groups, in performing tasks that primarily depend on efficient coordination.

A final group of studies reinforces the conclusion that hierarchical organization is important for coordination. These are experiments in which the communication network in a group is manipulated. Research by Leavitt, based on earlier work by Bavelas, imposed various communication networks on groups of five subjects.[21] A variety of networks was used, but we shall consider only the two extremes of centralization and decentralization. In the "wheel" or "X," a central person could communicate with all others, and they could communicate only with him. In the "circle" or "O," every person communicated with two neighbors. All subjects sat in separate cubicles and communicated in writing only. Each subject was given a sheet with five symbols on it and his task was to determine which one symbol was common to all the sheets. This is essentially a problem of coordination. Although there is a single correct solution, the difficulty is bringing the appropriate pieces of information together; once they are brought together, arriving at the answer is a trivial problem. The performance of wheel groups has been consistently found to be superior to that of circle groups. A centralized network of communication apparently contributes to effective coordination.

Later studies suggest that the crucial factor is not the formal network of communication but the group's ability to become hierarchically organized. The wheel imposes such hierarchical organization on a group, since the person in the central position naturally assumes the dominant role of coordinator. But if other types of groups are able to achieve status differentiation, then their performance becomes as good as that of the wheel groups. For example, Guetzkow and Simon, using the same equipment and task as Bavelas and Leavitt, compared the performance of groups when members were linked by the wheel, the circle, and an all-channel network (all members can communicate with all others).[22] Previous calculations had indicated that there were no significant differences between the three networks in terms of the time needed to accomplish the task *if* they were efficiently utilized;

[21] Harold J. Leavitt, "Some Effects of Certain Communication Patterns on Group Performance," *Journal of Abnormal and Social Psychology,* 46 (1951), pp. 38–50; see also, Alex Bavelas, "Communication Patterns in Task-Oriented Groups," *Journal of the Acoustical Society of America,* 22 (1950), pp. 725–730.

[22] Harold Guetzkow and Herbert A. Simon, "The Impact of Certain Communication Nets upon Organization and Performance in Task-Oriented Groups," *Management Science,* 1 (1955), pp. 233–250. See also Maul Mulder, "Communication Structure, Decision Structure and Group Performance," *Sociometry,* 23 (1960), pp. 1–14.

that is, any differences among the groups using the three networks was not due to the intrinsic limitations of a given network but rather to the use made of it. However, the difficulty of the organizational problem varied from network to network. The circle groups were expected to have the most difficult organizational problem, since they needed to establish relays for message passing as well as select one member to coordinate and distribute information. The wheel groups faced the least difficult organizational problems, since the network imposed a structure upon them. All-channel groups were considered intermediate, their problems being to eliminate excess channels and decide upon one person as coordinator. After a trial period, 56 groups were given 20 successive tasks with two-minute breaks between them during which members were allowed to discuss ways of improving their organization (also by an exchange of written messages). Using as a measure of efficiency the time needed to complete the task, wheel groups reached their maximum efficiency after only a few trials, all-channel groups were slower during early trials but most eventually performed as well as wheel groups, and circle groups generally did not reach optimum performance during the 20 trials. When and whether all-channel and circle groups achieved maximum efficiency depended on their success in developing a differentiated structure for communication.

In short, performance was negatively correlated with the difficulty of the organizational problems facing the groups. Among the groups of each type that did develop a hierarchical organization,[23] however, there were no significant differences in the efficiency with which they performed their tasks. A later publication analyzed the messages sent by group members during the periods between trials and found that whether or not the all-channel or circle groups developed a hierarchical organization was related to the number of specific proposals made and promulgated in planning such organizational arrangements.[24]

These experimental results strengthen our previous conclusions that hierarchical organization serves important functions for achiev-

[23] All 15 wheel groups, 17 out of 20 all-channel groups, and only 3 of 21 circle groups developed hierarchies during the 20 trials.

[24] Harold Guetzkow and William R. Dill, "Factors in the Organizational Development of Task-Oriented Groups," *Sociometry*, 20 (1957), pp. 175–204. Guetzkow and Dill also tested the hypothesis that circle groups would be more likely to achieve organization if they were permitted all-channel communication between trials. The results, however, were negative: the circle groups that were permitted intertrial all-channel communication were slightly more likely to become organized than regular circle groups but the average performance of the two types of groups was similar because the performance of the unorganized groups among those with all-channel networks between trials was particularly low.

ing coordination and that it does so specifically by restricting the free flow of communication. There is also some support for our hypothesis that the free flow of communication without hierarchical or other barriers is best suited for advancing new solutions to problems, including the ideas necessary to create a hierarchical organization where none exists. While hierarchically organized groups performed the coordination task better than groups not so organized, all-channel groups were more successful in developing an effective organization than were those groups where communication was restricted (circle groups).

Field Studies of Communication in Formal Organizations

Consultation among Peers. In turning from a discussion of controlled experiments to field studies concerned with communication, we are leaving behind the precision and safety of the laboratory. However, only by entering the field can we ascertain whether conclusions arrived at in the laboratory also hold for natural groups. One question is how accurately the experiment simulates the conditions it is supposed to represent, say, how well laboratory groups whose members are told they have similar interests reflect the salient aspects of solidarity in actual work groups. An even more fundamental problem is posed by the fact that experiments are designed to determine how a given factor in pure form and in the absence of variations in other conditions influences a second factor. Outside the laboratory, however, factors do not occur in pure form (there is no absolute vacuum, and neither is there completely unrestricted communication). Whether a condition occurs, or what effect it has if it does occur, is not independent of but contingent on other existing conditions. Field studies are needed to ascertain which social conditions typically occur together and which combinations of factors may be relevant for producing a given effect. Experiments are required to confirm the hypotheses derived from such field studies that a certain factor or set of factors actually has a certain effect. And field studies, in turn, must validate generalizations derived from experiments by showing that they apply outside the laboratory. The best approach, then, is to alternate between both methods; but in most areas of social research, experimentation is rarely used and, indeed, may not be feasible. Communication is one of the few subjects that have been investigated experimentally as well as in field studies and surveys, and this double-pronged approach lends it special significance.

As our first field study on communication, we undertake to analyze the interaction processes among colleagues in a federal law-enforcement agency.[25] The main duty of the agents was to investigate business establishments in order to determine their compliance with two federal laws. The work was complex and involved a high degree of discretion. Agents not only had to know—or at least to know about—the many general regulations that governed their work, including legislative acts and amendments, administrative interpretations, and court decisions, but they also had to be able to gather and sift information from many sources on the firms investigated. The duty to detect possibly concealed violations as well as the complexity of the task required that agents be able to exercise discretion; uniform law enforcement, however, required that decisions conform strictly to legal principles. The control system in the organization was designed to meet this double requirement.

The operations of the agents were controlled through an evaluation of results achieved rather than by means of detailed operating rules. The agents enjoyed considerable freedom in deciding how to proceed—for example, nobody checked on how they spent their time either in or out of the office—but their results (decisions) had to conform strictly to the law. They knew that their work required rigid conformity to the law of the land since making exceptions, even for the sake of adhering to the intent of the law, could open a Pandora's box of legal loopholes. The importance of attaining correct results in strict conformity with the legal standards was emphasized by having cases reviewed by both the supervisors and a special review unit, and statistical records of performance also focused attention on results; for example, records were kept on the proportion of cases in which violations had been discovered. Errors of decision detected by either review and the quantitative performance record were given great weight in the periodic rating of agents by their supervisors.

Indications were that evaluation on the basis of results achieved rather than methods employed fostered disciplined responsibility among agents. Emphasis in an organization on conformity with operating procedures discourages the exercise of initiative and the willingness to assume responsibility. The federal-agency system, in contrast, permitted individual agents considerable freedom in their work but exerted constraints to make their final decisions conform to general legal principles. Not the road taken but the destination reached was the test of performance. This reward system promoted responsibility among agents in both senses of the term: it encouraged them to assume responsibility for their decisions and to exercise discretion, and it also

[25] See Blau, *op. cit.*, pp. 99–115.

held them accountable for making responsible decisions in conformity with all the relevant official standards.[26]

Because agents were free to arrive at their own decisions, their work satisfaction was high. The discretion they exercised made their jobs challenging and interesting. But such responsibility engendered anxieties over decision-making which interfered with the capacity to make correct decisions. If agents encountered problems in their work, they were expected to confer with their supervisors, who might in turn send them to a staff attorney for further advice. However, this official procedure did not and could not lessen the anxiety aroused by concern over the correctness of decisions. Since the supervisor's evaluation of their work was a major cause of this anxiety, agents could not relieve it by coming to him with their difficulties. In fact, agents were reluctant to go to a supervisor with their problems for fear that doing so would reveal their ignorance—or, at least, their inability to act independently—and thus have adverse effects on their ratings.

These obstacles to the use of official channels of communication led agents to satisfy their needs for advice and social support by consulting one another. While the practice of turning to colleagues for advice was officially prohibited, it was tolerated, inasmuch as such operating rules were generally not strictly enforced. Consultation among peers appears to have had important consequences for the organization. This cooperative practice transformed an aggregate of individuals who merely had the same supervisor into a cohesive group. Moreover, it improved decision-making, not only by supplying information and advice when needed but also by reducing anxiety over decisions generally. The mere knowledge that one could ask a colleague for help when in difficulty decreased the anxiety engendered by the need to make complex decisions. Besides, the experience of being regularly asked by others for advice increased an agent's confidence in his own decisions. Hence, the decision-making ability of both participants was likely to benefit from consultations. By reducing anxiety and strengthening self-confidence, the pattern of unofficial consultation probably improved decisions even when no consultation took place. It should be noted that this conclusion is inferential. Although participation in consultation was associated with superior performance,[27] it is impossible to tell whether this association actually indicates that participation improved performance or merely that competent agents were more apt to be drawn into consultation. The latter was undoubtedly the case, but the question is whether the former was

[26] On the double meaning of "responsibility," see Marshall E. Dimock, *Administrative Vitality*, New York: Harper, 1959, pp. 218–219.

[27] See Blau, *op. cit.*, pp. 106–107.

also the case. To demonstrate that consultation affects decision-making would require an experimental or panel design.

Hierarchical status differences created obstacles to communication in this field situation, just as they did in laboratory experiments. Other field studies provide further support for the validity of this generalization. Argyris reports that supervisors in an industrial setting were very selective in the information they communicated to their superiors, tending to minimize problems, to emphasize successes, and to relay information detrimental to other supervisors whenever possible.[28] And research conducted by Blau in an employment agency indicates how operating directives became adjusted, redefined, and amplified as they passed from the upper levels of the hierarchy to the lower.[29] Finally, a study by Zaleznik replicates the finding from the law-enforcement agency in a machine shop.[30] These manual workers were also reluctant to approach their supervisor with their problems and instead turned to their peers. In particular, workers focused their requests for help on one individual who was high in seniority and who appeared to function as an informal leader.

The Dynamics of Communication Processes. A case study in which Shepard analyzes the patterns of consultation in a university research organization provides some interesting contrasts and parallels.[31] Most of the staff members of this engineering laboratory were recent university graduates who held temporary positions, which they looked upon primarily as a valuable opportunity to acquire good research experience—an advanced program of postgraduate training. Salaries were low compared with those engineers could command in industry. The staff was willing to sacrifice present income for experience and anticipated future earnings. Salaries were determined by the university, and superiors in the research organization had no control over them. Moreover, there was little chance for (and little interest in) promotion. Hence, employees were not dependent upon their superiors for receiving formal rewards—a situation quite different from that in most organizations.

We have seen that hierarchical dependence blocks the free flow of communications and, more specifically, disinclines staff members to

[28] Chris Argyris, *Executive Leadership,* New York: Harper, 1953, pp. 46–48.

[29] Blau, *op. cit.,* pp. 21–28.

[30] Abraham Zaleznik, *Worker Satisfaction and Development,* Boston: Graduate School of Business Administration, Harvard University, 1956, pp. 36–41.

[31] Herbert A. Shepard, "The Value System of a University Research Group," *American Sociological Review,* 19 (1954), pp. 456–462. For a study that compares communication patterns in the administrative and the scientific sections of a large federal research bureau, see Robert S. Weiss, *Processes of Organization,* Ann Arbor: Institute for Social Research, University of Michigan, 1956.

discuss their problems with a superior for fear of revealing their ignorance to him. Where creativity is valued there is also the danger that free discussions will not occur even among peers, since each person will be concerned with "getting credit" for his new ideas. Such communication barriers were not prevalent in this research organization: engineers readily discussed their problems with one another regardless of official rank.[32] Although he had no quantitative data it was Shepard's impression that this practice of freely talking problems over with colleagues stimulated ideas and improved research productivity. Indeed, these discussions seem to have served as a means for acquiring status and as an informal reward system. In the absence of formal rewards the prestige gained by demonstrating technical knowledge and solving complex problems provided an incentive for exerting effort. Moreover, the exchange of technical ideas in these discussions was a learning experience for all participants, which constituted another informal reward.

Let us contrast communication patterns in the university research group with those found in the federal law-enforcement agency. The free flow of consultations among colleagues probably improved performance in both organizations. Communications across hierarchical lines were more restricted in the law-enforcement agency than in the research organization—an expected consequence of the lesser hierarchical dependence in the latter. As we have seen, however, hierarchical dependence did not appear to impede consultation among colleagues in the federal agency. Perhaps advice would have been given less freely if the agents in the federal agency had received formal credit for having original ideas. But they obtained formal rewards only for ideas that contributed to their own cases, not for all good ideas, such as those that might help solve the problems in a colleague's cases. In other words, federal agents were engaged in similar but not common work, and ideas were given official recognition only in relation to particular case problems. Since these agents, just as Shepard's engineers but for different reasons, could not gain formal rewards by withholding ideas from colleagues, the informal rewards they could gain from freely giving advice governed their conduct.

Shepard suggests that in the research organization, earning the respect of colleagues served as a substitute for formal rewards. But if the unofficial pattern of rewarding competence in problem-solving with high respect has the same function for performance as the official institution of rewarding such competence with good ratings and ad-

32 There did appear to be some communication barriers between engineers and technicians (less highly trained employees). Shepard felt (*op. cit.*, p. 461) that these stemmed from the difficulties of reciprocation: technicians could learn much from engineers but not engineers from technicians.

vancements, may the two not also have the same dysfunctions? If the informal status structure constitutes an effective incentive system, just as the formal status structure does, one would expect differences in informal status to impede communications, just as differences in formal status do. Fear of losing face is probably not much less inhibiting than fear of losing one's place. The study of the research organization provides no information on this matter, but the study of the federal agency does.[33] The most competent agents were most often consulted, but most agents discussed the majority of their problems not with these experts but with others of lesser competence. Agents explained this preference by saying that they were reluctant to go regularly to other agents with whom they were not particularly friendly. Instead, they tended to establish partnerships for the purpose of mutual consultation.

Why did agents settle for advice which they knew to be not the best available? A person receives advice in exchange for paying respect. By repeatedly approaching another for advice, the questioner is implicitly acknowledging the superior competence of the adviser. The consultant earns respect, in exchange for which he must devote time and energy to helping colleagues with their problems. However, this system undergoes modifications that can be looked upon as resulting from marginal-utility functions. The respect received for any single consultation declines as the number of consultations increases. A person seldom consulted will appreciate the respect for his judgment shown by a colleague who asks for his advice much more than the one often consulted. Correspondingly, the expenditure of time becomes increasingly difficult to bear as frequent consultations infringe more and more on the time a colleague needs for work on his own cases. Consequently, popular consultants find requests for advice less welcome than others, and may even become impatient and uncooperative when colleagues ask too frequently for their help. The cost of advice to the one who requests it changes in similar ways. Continually paying respect and giving deference to colleagues by asking for their help threatens the self-confidence of an agent and his standing among peers. If the cost of obtaining expert advice becomes too high for an agent, partly because he needs so much of it, he will seek to obtain advice of inferior quality at a cheaper price.

These processes promote the establishment of partnerships for mutual consultation, particularly among officials in frequent need of assistance. No deference is implied by asking the advice of a colleague who often asks one's own advice. In these partnerships, therefore, agents paid for advice not with deference, which they could ill afford, but with consulting time devoted to the partner's problems. This was

33 See Blau, *op. cit.*, pp. 106–110.

a cheap price, since little demand was made on the consulting time of the less competent agent and he enjoyed being asked for advice. Inasmuch as the anxiety aroused by having to submit decisions to superiors for evaluation interfered with decision-making, agents could benefit from consulting a colleague whose competence was not superior to their own, for *his* decision-making ability on *their* cases was not impeded by such anxiety. Furthermore, the development of consulting partnerships enabled agents to make fewer demands on the time of the experts and thus to approach them more freely when faced with a particularly difficult problem.

Let us summarize the implications of this analysis. The formal status hierarchy in an organization creates obstacles to the free flow of communication. Specifically, dependence on superiors for formal rewards restricts consultation across hierarchical boundaries. Under special conditions, it may also discourage consultation among colleagues, but probably the more typical situation is that hierarchical obstacles to communication foster consultation among peers. The processes of consultation among peers give rise to an informal differentiation of status, because some members of the colleague group earn more respect as consultants than others. Such emerging distinctions of informal status also create obstacles to the free discussion of problems, just as formal status differences do. These obstacles may further redirect the flow of consultation, so that the highest frequency occurs between persons of equal informal as well as formal status. But even consultations among persons of equal competence probably improve the quality of performance, for anxiety interferes less with making decisions on a colleague's cases than with making decisions on one's own.

Variations in Communication Patterns

Reciprocity in Consultation. The foregoing discussion has shown that whether consultations are reciprocal or one-sided has important consequences for the participants and the group structure. Unilateral consultations, where a given worker usually asks another for help, create a differentiation of informal status. By recurrently requesting a colleague's advice, a person socially acknowledges that the other's standing as an expert is superior to his own. Moreover, receiving advice creates social obligations, which constrain a person to return the favor by deferring to his consultant's wishes and suggestions. Hence, power as well as prestige becomes differentiated as the result of one-sided consultations. In unilateral consultations advice is exchanged for informal standing, while in consulting partnerships there is a